

California Housing Price Prediction

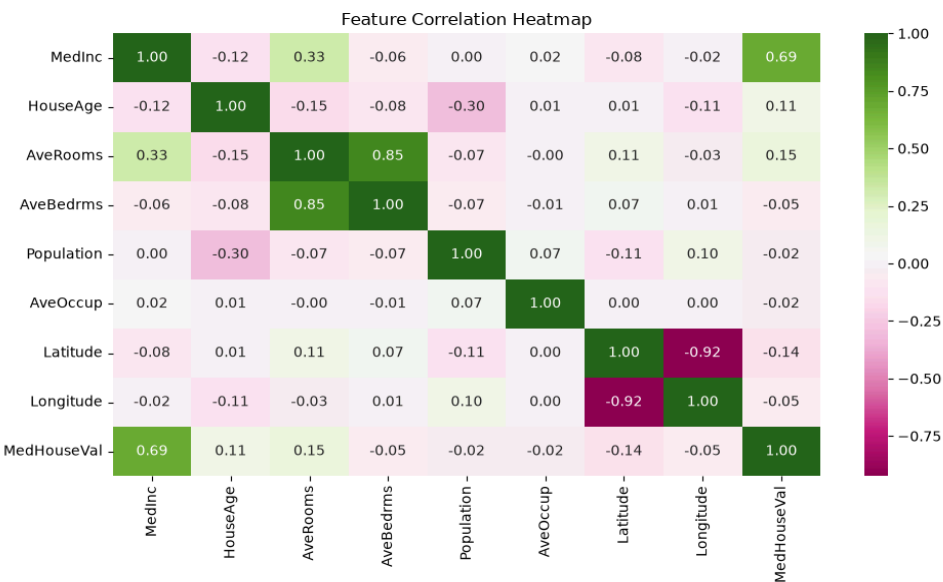
Executive Summary & Objective

This project is developed to implement an end-to-end Machine Learning workflow from data acquisition and exploratory data analysis (EDA) to model training, evaluation, and diagnostic error analysis. The California Housing dataset was utilized to build an Ordinary Least Squares (OLS) Linear Regression model predicting median house values across California census block groups.

The baseline linear model demonstrates a statistically significant ability to capture housing valuation trends, explaining approximately 57.6% of target variance ($R^2 = 0.5757$). This report details dataset distributions, correlation dynamics, regression performance benchmarks, residual diagnosis, and concrete architectural strategies for model refinement.

Exploratory Data Analysis (EDA)

Total Observations -	20,640
Feature Columns -	8 continuous
Missing / Null Values -	0 (Clean)
Target Variable -	MedHouseVal (\$100k)
Target Range -	\$0.150 to \$5.000
Target Mean -	\$2.068 ± \$1.154



Key Statistical Insights:

- Primary Driver (MedInc):** Median Income exhibits the highest positive linear correlation with house values ($r \approx 0.69$), establishing household income as the dominant linear predictor.
- Multicollinearity:** High pairwise correlation exists between AveRooms and AveBedrms ($r \approx 0.85$), introducing slight variance inflation in linear weight estimation.
- Target Saturation:** The dataset has an artificial price ceiling capped at \$5.000 (\$500,000), producing boundary anomalies.

Model Methodology & Implementation

To evaluate generalization capability and avoid data leakage, the full dataset (20,640 samples) was partitioned into an 80% training set (16,512 samples) and a 20% test set (4,128 samples) using `train_test_split` with `random_state=42` for deterministic reproducibility.

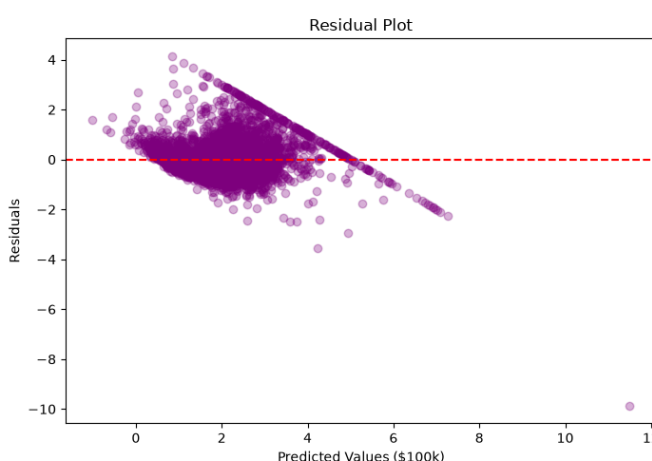
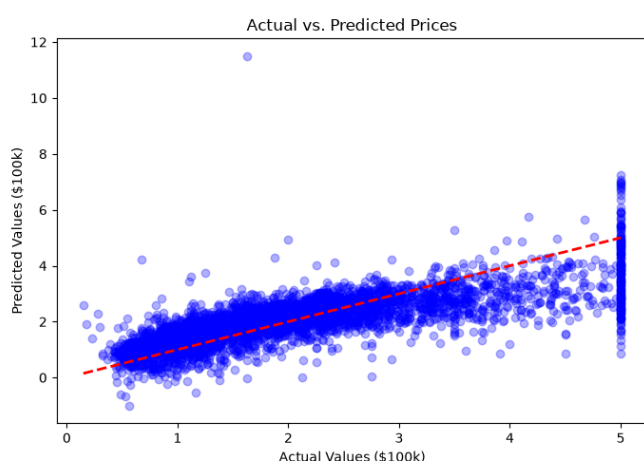
An Ordinary Least Squares (OLS) Linear Regression model was fitted on the training split `X_train`, learning regression coefficients (weights) that minimize the sum of squared residuals between observed and predicted values:

$$y_{pred} = \beta_0 + \beta_1(MedInc) + \beta_2(HouseAge) + \beta_3(AveRooms) + \beta_4(AveBedrms) + \beta_5(Population) + \beta_6(AveOccup) + \beta_7(Latitude) + \beta_8(Longitude)$$

Evaluation & Performance Metrics

The trained model was evaluated on the held-out 20% test set (4,128 unseen samples). Performance was assessed using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the Coefficient of Determination (R^2).

Metric	Formula	Score	Interpretation
MAE (Mean Absolute Error)	$\frac{1}{n} \sum y_i - \hat{y}_i $	0.5332	Interpretation Average absolute deviation between predicted and actual prices is ~\$53,320
RMSE (Root Mean Squared Error)	$\sqrt{\frac{1}{n} \sum (y_i - \hat{y}_i)^2}$	0.7455	Penalizes larger outliers more heavily; useful for measuring extreme valuation errors
R^2 Score (Coefficient of Determination)	$1 - (SS_{res} / SS_{tot})$	0.5757	Baseline OLS explains ~57.6% of the total variance present in the test set



Diagnostic Analysis

- Actual vs. Predicted: Strong agreement along the 1:1 diagonal for median prices between \$100k and \$350k. Clear under-prediction for luxury properties capped at \$500k.
- Residual Distribution: Residuals are centered near zero with slight heteroscedasticity for high-value properties due to the dataset ceiling cap.

Limitations & Strategic Improvement Roadmap

While the baseline Linear Regression model serves as an effective benchmark, several architectural and data-

level enhancements can substantially improve predictive accuracy (targeting $R^2 \geq 0.82$):

1. **Skew Handling & Transformation:** Features like AveRooms, AveBedrms, Population, and AveOccup exhibit heavy right-skewed distributions. Applying log transformations (np.log1p) or QuantileTransformer will normalize input scales and mitigate outlier disruption.
2. **Geographic Distance Feature Engineering:** Linear regression interprets Latitude and Longitude independently, failing to model spatial proximity. Engineering geodesic distance features to major coastal economic hubs (e.g., San Francisco, Silicon Valley, Los Angeles, San Diego) will capture localized market premiums.
3. **Ratio & Interaction Terms:** Constructing domain-specific interaction features—such as $\text{Rooms_per_Household} = \text{AveRooms} / \text{AveOccup}$ and $\text{Bedrooms_per_Room} = \text{AveBedrms} / \text{AveRooms}$ —will expose non-linear relationships to linear estimators.
4. **Regularization (Ridge / Lasso / ElasticNet):** Applying L1 (Lasso) and L2 (Ridge) penalties will address the multicollinearity observed between AveRooms and AveBedrms, stabilizing model weights and suppressing uninformative variance.
5. **Non-Linear Tree Ensembles:** Transitioning to tree-based ensemble architectures (Random Forest, LightGBM, XGBoost) will natively partition non-linear geographical boundaries and complex interaction terms, drastically reducing both MAE and RMSE.

Conclusion

This task successfully validates the ML lifecycle on the California Housing dataset, establishing an interpretable baseline ($R^2 = 0.576$, $\text{MAE} = 0.533$). All exploratory data analyses (EDA), correlation heatmaps, model training routines, metric computations, and residual diagnostic visualizations are fully documented and executable within the primary deliverable notebook:

task1_ml_linear_regression.ipynb.